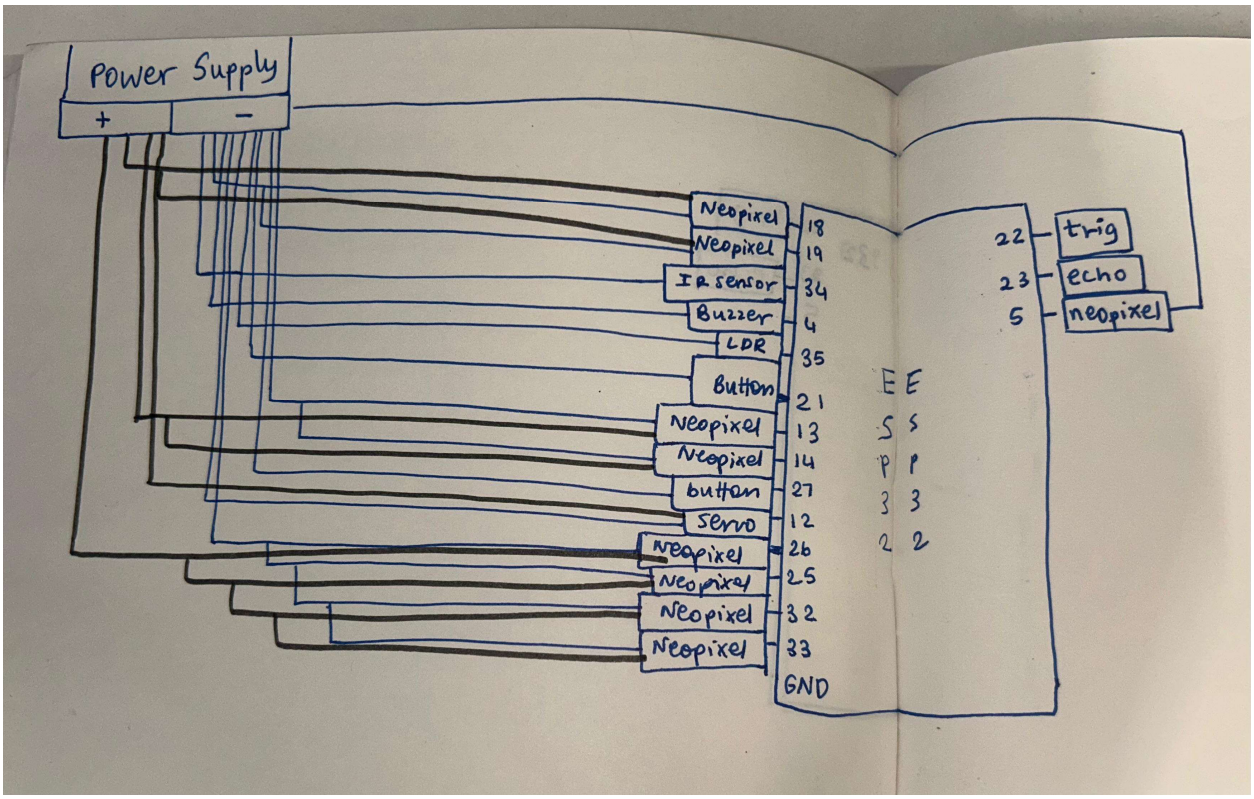
Electronics Documentation

1. Electronics Block Diagram



2. Components Table

Component	Quantity	Purpose	Justification
ESP32 microcontroller	1	Runs all game logic, manages all states, handles BLE	required for Trial 2 MIT app interface and sufficient GPIO count for the full pin requirements
Neopixel strip (gobring, 16 LED)	1	Shows ambient light and various	Individually addressable LEDs allow different animations and brightness modifications and can be controlled by one GPIO

		stages of the game	Pin instead of multiple pins like with leds.
Neopixel strip (idle strip, 10 LED)	1	Visual feedback during idle state and IR detection confirmation	Individually addressable LEDs allow different animations and brightness modifications and can be controlled by one GPIO Pin instead of multiple pins like with leds.
Neopixel strip (dragon body, 10 LED)	1	Moving heartbeat pulse in Trial 1	Moving light effect requires individual pixel control, which is only achievable with individually addressable LEDs
Neopixel strip (eye strip, 9 LED)	1	Dragon eye animation. flashes red on wrong press,	Individually addressable LEDs allow different animations and brightness modifications and can be controlled by one GPIO Pin instead of multiple pins like with leds.
Neopixel strip (sequence display, 17 LED)	1	Displays the 6 symbol sequence in Trial 2 using LED pairs to represent each symbol	Individually addressable LEDs allow different animations and brightness modifications and can be controlled by one GPIO Pin instead of multiple pins like with leds.
Neopixel strip (oxygen cylinder, 10 LED)	1	Depleting oxygen level indicator in Trial 3 ,it fills from bottom, depletes upward over time	Progressive drain effect requires individual pixel control, which is only achievable with individually addressable LEDs

Neopixel strip (harvest bed, 6 LED)	1	Three active LEDs in the gillyweed bed show remaining harvests. green means remaining, turns off when harvested, flashes red on wrong attempt	Individually addressable LEDs allow different animations and brightness modifications and can be controlled by one GPIO Pin instead of multiple pins like with leds.
Neopixel strip (target indicator, 10 LED)	1	Shows which height the player needs to estimate in Trial 3 ,one pixel lights at the target position on the strip	Individually addressable LEDs allow different animations and brightness modifications and can be controlled by one GPIO Pin instead of multiple pins like with leds.
Neopixel strip (trophy rainbow, 16 LED)	1	Victory animation	Rainbow animation requires requires individual pixel control, which is only achievable with individually addressable LEDs
Servo motor (dragon mouth)	1	Pulls string attached to dragon jaw. opens the mouth on Trial 1 win to reveal the egg	Servo allows a controlled gradual open motion rather than a sudden snap and fits within 5V supply limits
Ultrasonic sensor	1	Measures distance of player's estimate in Trial 3 gillyweed harvest	Gives distance readings without contact. Requires only 2 GPIO pins .Readings averaged over multiple samples for stability

IR sensor	1	Detects paper drop in goblet bowl during idle state to trigger game start	IR would be more reliable than a ldr sensor in this case since its placed inside the goblet.
LDR (egg holder)	1	Detects when egg is placed in holder	Takes input when light is covered, ideal for an egg holder
Buzzer	1	Audio feedback for elimination	Requires only one GPIO pin and is sufficient for an elimination buzzer
Push button (trial 1)	1	Player input for the heartbeat stop mechanic in Trial 1	Physical button would provide accurate readings appropriate for reaction time
Push button (transition from T2 to T3)	1	Player presses to start Trial 3	Button ensures player is ready , avoids auto triggering of game.
Breadboard power supply	1	Powers all the neopixels, sensors, servos etc	A power supply prevents voltage fluctuations that could cause neopixel colour errors or ESP32 resets. (one supply failed during development at 12V output)